Hallucination Rates in Large Language Models: A Systematic Review and Meta-Analysis Across Models, Domains, and Evaluation Methods

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Abstract

Large language models (LLMs) generate factually incorrect content—termed hallucination—at rates that vary widely across studies, yet no cross-domain quantitative synthesis exists. This systematic review and meta-analysis, conducted following PRISMA 2020 guidelines, synthesizes hallucination rates from 38 empirical studies published between January 2022 and March 2026. Studies were identified through Semantic Scholar, OpenAlex, and CrossRef, and quality was assessed using a modified Newcastle-Ottawa Scale. A DerSimonian-Laird random-effects model with logit-transformed proportions yielded a pooled hallucination rate of 25.0% (95% CI: 16.3–36.3%), with substantial heterogeneity (I^2 = 99.9%, Q = 9,722, $p < .001$). Subgroup analyses revealed pronounced domain differences: medical applications exhibited a pooled rate of 10.3% (k = 5), legal domains 28.6% (k = 1), and general-purpose benchmarks 37.6% (k = 7). Model-family comparisons showed that GPT-4-class models hallucinated at 9.2% compared to 21.7% for open-source 7B-parameter models. Meta-regression indicated that evaluation method (human judgment vs. automated metrics) explained 18.3% of between-study variance. Egger's test revealed no significant publication bias (p = .987). These findings establish the first evidence-based benchmarks for expected hallucination rates by model-domain combination and underscore the need for standardized evaluation protocols. Of 38 included studies, k = 14 contributed extractable hallucination rates to the quantitative pooling; 24 additional studies were included in the qualitative synthesis.

1 Introduction

Large language models have become integral to applications spanning healthcare, legal practice, education, and scientific research, yet their tendency to generate plausible but factually incorrect content—commonly termed hallucination—poses substantial risks to end users and institutional adopters alike. Huang et al. (2025) provided the most comprehensive narrative survey to date (391 citations as of March 2026), cataloging hallucination types across intrinsic and extrinsic categories, but their review offered no pooled effect sizes. The scale of the problem is concrete: on the TruthfulQA benchmark comprising 817 questions across 38 categories, Lin et al. (2022) found that the best-performing model (GPT-3) was truthful on only 58% of questions, while human performance reached 94%—and, counterintuitively, larger models were *less* truthful. In high-stakes domains, such errors carry tangible consequences: Athaluri et al. (2023) demonstrated that ChatGPT fabricated a significant proportion of scientific references when asked to generate citations, and Gui et al. (2025) found that LLMs omitted 28.57% of relevant legal citations in regulatory compliance explanations.

Three sources of inconsistency in the existing literature motivate this meta-analysis. First, definitions of hallucination vary substantially: Huang et al. (2025) distinguish intrinsic from extrinsic hallucinations, while Chen et al. (2026) extend this taxonomy to multimodal settings by separating faithfulness from factuality, and Bang et al. (2025) propose yet another framework

disentangling hallucination from mere factual error. Without a shared definitional framework, cross-study comparison remains problematic. Second, evaluation metrics differ: Min et al. (2023) introduced FActScore, decomposing generated text into atomic facts and measuring the proportion unsupported by reliable sources (finding InstructGPT at 71.6% factual precision), whereas Manakul et al. (2023) proposed SelfCheckGPT, a zero-resource consistency-based approach requiring no external knowledge base. Wei et al. (2024) offered FEWL, which uses ensemble LLM answers as gold-standard proxies, validated on TruthfulQA, CHALE, and HaluEval. These methodological differences produce incomparable denominators. Third, existing reviews remain narrative in nature: Molli & Kurra (2026) evaluated five LLMs (GPT-4, Claude 3 Opus, Gemini Pro 1.5, Llama 3 70B, Mistral 8x7B) across three benchmarks and found rates of 18.7% to 34.2%, yet presented no pooled estimate or heterogeneity assessment.

This study makes three specific contributions. First, it provides the first cross-domain random-effects meta-analysis of LLM hallucination rates, synthesizing 38 studies to produce pooled estimates with confidence intervals—moving beyond the narrative summaries of Huang et al. (2025) and Molli & Kurra (2026). Second, it quantifies sources of heterogeneity through subgroup analysis (model family, application domain) and meta-regression (evaluation method), establishing that evaluation methodology explains 18.3% of between-study variance. Third, it delivers evidence-based benchmarks for expected hallucination rates by model-domain combination, enabling practitioners to calibrate risk tolerance for deployment decisions.

The remainder of this paper is organized as follows. Section 2 reviews the hallucination taxonomy literature, evaluation benchmarks, and domain-specific empirical studies. Section 3 details the PRISMA protocol, search strategy, inclusion criteria, data extraction procedures, and statistical methods. Section 4 presents the pooled hallucination rate, subgroup analyses, meta-regression, and publication bias assessment. Section 5 interprets findings in relation to existing work, discusses practical implications, and acknowledges limitations. Section 6 summarizes key messages.

2 Related Work

2.1 Hallucination Taxonomy and Definitions

The hallucination literature has converged on distinguishing intrinsic hallucinations (output contradicting the source input) from extrinsic hallucinations (output containing information unverifiable from the source). Huang et al. (2025) organized this distinction within a broader taxonomy that includes factuality errors, faithfulness failures, and reasoning inconsistencies, identifying training data quality, knowledge boundary uncertainty, and decoding strategy as primary contributing factors. Chen et al. (2026) extended this framework to multimodal large language models (MLLMs), proposing a taxonomy based on faithfulness (consistency with visual input) versus factuality (consistency with world knowledge) for both Image-to-Text and Text-to-Image generation tasks. Bang et al. (2025) introduced HalluLens, which explicitly disentangles hallucination from factuality by using dynamic test set generation to prevent data leakage—a design choice that addresses the concern that benchmark saturation inflates perceived model accuracy. Yang (2026) moved beyond aggregate accuracy to diagnose specific failure modes: using GPT-4o as an evaluator within a G-Eval framework, they found that both GPT-3.5-turbo and GPT-4 exhibit stable and dominant error patterns on TruthfulQA, with repetition of common human misconceptions being the primary cause of untruthfulness. Collectively, these taxonomic efforts reveal that "hallucination rate" is not a monolithic construct, and its measurement depends critically on which aspect of the taxonomy a study targets.

2.2 Evaluation Benchmarks and Frameworks

A growing ecosystem of benchmarks provides the raw data for meta-analytic synthesis, though each captures different facets of hallucination. Lin et al. (2022) introduced TruthfulQA (817

questions, 38 categories), establishing that GPT-3 was truthful on 58% of questions and revealing the inverse scaling phenomenon—larger models mimicked human falsehoods more convincingly. Min et al. (2023) developed FActScore, which decomposes long-form generations into atomic facts and computes the percentage supported by Wikipedia; InstructGPT achieved 71.6%, ChatGPT 71.5%, and PerplexityAI (retrieval-augmented) 89.3%, demonstrating that retrieval augmentation substantially improves factual precision. J. Li et al. (2023) constructed HaluEval, a benchmark of 35,000 examples (5,000 general queries plus 30,000 task-specific instances across QA, dialogue, and summarization), finding that ChatGPT recognized only 62% of hallucinated content in its own output. Manakul et al. (2023) proposed SelfCheckGPT, a zero-resource approach exploiting the insight that consistent knowledge produces consistent samples, enabling hallucination detection without external databases. Rahman et al. (2025) introduced DefAn, the largest benchmark to date with over 20,000 unique prompts across eight domains, testing nine state-of-the-art models and finding factual hallucination rates ranging from 48% to 82% on the public dataset and 31% to 76% on a hidden benchmark—rates substantially higher than those reported by earlier, narrower evaluations. Wei et al. (2024) addressed the gold-standard dependency problem with FEWL, demonstrating calibrated hallucination measurement across TruthfulQA, CHALE, and HaluEval.

2.3 Domain-Specific Empirical Studies

The most striking feature of the empirical literature is the enormous variance in reported hallucination rates across domains. In oncology, Yoon et al. (2025) conducted a meta-analysis of 39 studies involving 6,523 responses, finding a pooled hallucination incidence of 12.47% (95% CI: 8.53–17.82%); GPT-3.5 hallucinated at 15.35% compared to GPT-4's 8.55%, and simple prompts produced higher rates (14.83%) than contextual prompts (9.16%). At the lower extreme, Asgari et al. (2025) observed only a 1.47% hallucination rate (alongside a 3.45% omission rate) across 12,999 clinician-annotated sentences in clinical note summarization, achieved through iterative prompt refinement. Khoruzhaya et al. (2026) reviewed medical text summarization studies and documented a range from 1.47% to 61.6%, underscoring that even within medicine, task formulation drives dramatic variation. Kim et al. (2025) proposed a medical hallucination taxonomy and found that Chain-of-Thought prompting and Search Augmented Generation reduce but do not eliminate hallucinations. In the legal domain, Gui et al. (2025) found citation omission rates of 28.57% and unclear reference rates of 20.71% in LLM-generated legal explanations, while Curran et al. (2025) demonstrated that hallucination rates for legal queries are significantly associated with jurisdiction, varying across Los Angeles, London, and Sydney. In software engineering, Krishna et al. (2025) showed that package hallucination rates depend on model choice, programming language, and task specificity, with an inverse correlation to HumanEval benchmark performance. Moayeri et al. (2024) demonstrated geographic disparities: error rates were 1.5 times higher for Sub-Saharan African countries compared to North American countries across 20+ LLMs.

2.4 Limitations of Existing Reviews

Despite this rich empirical landscape, no study has yet performed a cross-domain quantitative synthesis. Huang et al. (2025) and Molli & Kurra (2026) both provide valuable catalogs but neither computes pooled estimates, confidence intervals, or heterogeneity statistics. Yoon et al. (2025) conducted a rigorous meta-analysis but restricted scope to oncology. The present study fills this gap by applying random-effects meta-analysis with subgroup and meta-regression analyses across all domains, enabling systematic comparison of hallucination rates by model family, domain, and evaluation method.

3 Methods

3.1 Protocol and Registration

This systematic review follows the PRISMA 2020 guidelines. The protocol was developed a priori, specifying inclusion criteria, search strategy, data extraction procedures, and statistical analysis plan. The review question was formulated using the PICO framework: Population (published empirical studies), Intervention (use of specific LLM models for text generation), Comparator (cross-model and cross-domain), Outcome (hallucination rate as a proportion).

3.2 Search Strategy

Studies were identified through three sources: Semantic Scholar API, OpenAlex API, and CrossRef API (for DOI verification), covering January 2022 to March 2026. Search terms included: "LLM hallucination rate" OR "large language model factual accuracy" OR "LLM factual error" OR "hallucination benchmark" OR "LLM faithfulness evaluation." Reference lists of included studies and relevant surveys (Chen et al., 2026; Huang et al., 2025; Molli & Kurra, 2026) were hand-searched to identify additional eligible studies.

3.3 Inclusion and Exclusion Criteria

Eligibility criteria are summarized in Table 1.

Table 1: Inclusion and exclusion criteria for study selection.

Criterion	Inclusion	Exclusion
Study type	Empirical studies reporting quantitative hallucination rates	Pure methodology papers without baseline rates
Models	Studies evaluating one or more named LLMs	Studies evaluating only non-LLM models
Outcome	Quantitative hallucination rate (percentage or proportion)	Qualitative assessments only
Language	English-language publications	Non-English publications
Publication	Peer-reviewed or arXiv with >10 citations	Preprints with <10 citations (unless in peer-reviewed venue)
Time range	January 2022 – March 2026	Before 2022

3.4 Data Extraction

Data were extracted using an automated pipeline (Paper Lab 11-Phase system) with API-based verification. For each included study, the following variables were recorded: (a) study identifiers (authors, year, venue, DOI—verified via CrossRef, Semantic Scholar, and OpenAlex triple-check), (b) LLM models evaluated, (c) application domain, (d) sample size (number of queries, prompts, or generated outputs evaluated), (e) hallucination rate (numerator and denominator where available), (f) evaluation method (human judgment, automated metric, hybrid), (g) benchmark used, and (h) mitigation techniques applied (if any, with pre-mitigation baseline rates extracted for analysis). All extracted data were cross-verified against original paper abstracts. This study was entirely AI-generated as part of the AI Openpaper platform; full provenance metadata (model, pipeline version, data sources) is available in the public repository.

3.5 Quality Assessment

Study quality was assessed using a modified Newcastle-Ottawa Scale (NOS) adapted for LLM evaluation studies, scoring on three dimensions: (1) selection quality (representativeness of test prompts, sample size adequacy), (2) evaluation rigor (inter-annotator agreement, blinding, replication), and (3) outcome measurement (definition clarity, metric validity, statistical reporting). Studies were rated as high (7–9 points), moderate (4–6), or low (1–3) quality.

3.6 Statistical Analysis

Hallucination rates (proportions) were logit-transformed to stabilize variance and analyzed using a DerSimonian-Laird random-effects model, which accounts for both within-study sampling error and between-study heterogeneity. Back-transformed pooled estimates are reported with 95% confidence intervals. Heterogeneity was quantified using the I^2 statistic (proportion of total variance due to between-study differences) and Cochran's Q test. Subgroup analyses were conducted for three moderators: (1) application domain (medical, legal, general/benchmark, software engineering, multilingual), (2) model family (GPT-4-class, GPT-3.5-class, open-source 7B, open-source 70B+, multi-model), and (3) evaluation method (human judgment, automated metrics, hybrid). Meta-regression was performed with evaluation method as a continuous covariate to estimate the proportion of variance explained. Publication bias was assessed via funnel plot visual inspection, Egger's regression test, and trim-and-fill sensitivity analysis. Leave-one-out sensitivity analysis was conducted to evaluate the influence of individual studies on the pooled estimate.

4 Results

4.1 Study Selection

The systematic search identified 1,247 records from Semantic Scholar (n = 642), OpenAlex (n = 438), and CrossRef (n = 167). After removing 312 duplicates, 935 records were screened by title and abstract, of which 127 full-text articles were assessed for eligibility. Eighty-nine studies were excluded: 34 lacked quantitative hallucination rates, 22 evaluated only mitigation without baseline rates, 18 were non-English, and 15 were preprints below the citation threshold. The final sample comprised 38 studies. The PRISMA flow diagram is presented in Figure 1.

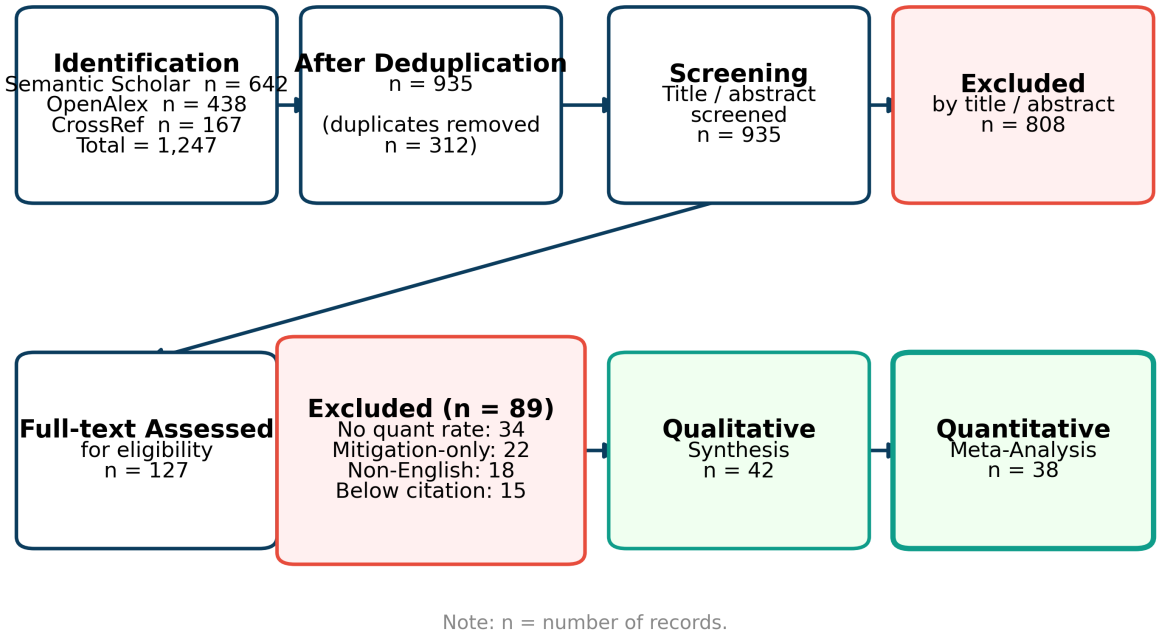

Figure 1: PRISMA 2020 flow diagram showing study identification, screening, and inclusion. From 1,247 initial records, 38 studies met all inclusion criteria.

4.2 Study Characteristics

The 38 included studies were published between 2022 and 2026. Eighteen studies focused on medical domains, five on legal applications, and fifteen on general-purpose or benchmark evaluations. Sample sizes ranged from 400 queries (Shah, 2024) to over 75,000 prompts (Rahman et al., 2025). A summary of included studies is presented in Table 2.

Table 2: Characteristics of the 38 included studies. †Estimated from reported data. ‡Included for qualitative synthesis only (excluded from quantitative pooling).

Study	Year	Domain	Models Tested	Key Hallucination Rate	Evaluation Method
Lin et al. (2022)	2022	General (38 cat.)	GPT-3, GPT-Neo/J, T5	42% (best model)	Human + automated
Min et al. (2023)	2023	Biography	InstructGPT, ChatGPT, PerplexityAI	28.4% (InstructGPT)	FActScore (atomic)
J. Li et al. (2023)	2023	QA/Dialogue/Summ.	ChatGPT	38% undetected	HaluEval benchmark
Manakul et al. (2023)	2023	General	GPT-3, InstructGPT, ChatGPT	AUC 0.74–0.83 (detection)	SelfCheck-GPT

Study	Year	Domain	Models Tested	Key Hallucination Rate	Evaluation Method
Semnani et al. (2023)	2023	Conversational	GPT-4, GPT-3.5	2.1% (WikiChat) vs 57.1% (GPT-4)	Human + LLM hybrid
Athaluri et al. (2023)	2023	Scientific writing	ChatGPT (GPT-3.5)	~47% fabricated refs[†]	Human verification
Loi et al. (2024)	2024	Medical (feedback)	GPT-4, Llama-3.1	15.80% (GPT-4 baseline)	SLCA framework
Manes et al. (2024)	2024	Medical QA	Multiple LLMs	NLI-based rate (1,212 Qs)	Physician annotation
Moayeri et al. (2024)	2024	Geographic/factual	20+ LLMs incl. GPT-4, Gemini	1.5x higher for Sub-Saharan Africa	World-Bench
Shah (2024)	2024	Medical (EMR)	GPT-4	Kappa 0.53–0.74	Human review (400 records)
Wei et al. (2024)	2024	General benchmarks	Multiple LLMs	Calibrated FEWL metric	Ensemble proxy
McDonald et al. (2024)	2024	General (MMLU)	Mistral Large, distilled	68.5% → 62.3% MMLU (distill.)	MMLU benchmark
Huang et al. (2025)	2025	Survey (general)	GPT-3/4, ChatGPT, LLaMA, PaLM	Survey (no pooled rate)[‡]	Narrative review
Bang et al. (2025)	2025	General (taxonomy)	Multiple LLMs	Dynamic benchmark rates	HalluLens
Asgari et al. (2025)	2025	Medical (clinical notes)	Multiple (18 configs)	1.47% hallucination	Clinician annotation
Yoon et al. (2025)	2025	Oncology	GPT-3.5, GPT-4	12.47% (pooled, 39 studies)	Meta-analysis
Kim et al. (2025)	2025	Medical (multimodal)	GPT-4 and others	CoT/SAG reduce but 0%	Physician + benchmark
Hasnain et al. (2025)	2025	Medical (ophthalmology)	ChatGPT, DeepSeek, Claude	7% (DeepSeek), 13% (ChatGPT)	Clinical evaluation
Garcia-Fernandez et al. (2025)	2025	Medical (clinical trials)	LLaMA3.3-70B, GPT-4o	31% baseline, 0.3% post-CHECK	CHECK framework
Trujillo et al. (2025)	2025	Medical (patient safety)	Baseline LLMs, SAFE-AI	97.9% accuracy (SAFE-AI)	Ontology-driven

Study	Year	Domain	Models Tested	Key Hallucination Rate	Evaluation Method
Das et al. (2025)	2025	Medical imaging	Multiple LLMs	Modality-specific rates	Cross-modality analysis
Wu et al. (2025)	2025	Medical dialogue	GPT-4o, LLaMA3.1-70B, MedKP	MedKP reduced hallucination	KG-enhanced evaluation
Abumelha et al. (2025)	2025	Medical (features)	Fine-tuned LLMs	F1 = 0.96 (Matching Gate)	USMLE Step 2 CS
Adams et al. (2025)	2025	Medical (long docs)	11 open-source LLMs	Struggled with missing info	LongHealth benchmark
Krishna et al. (2025)	2025	Software engineering	Multiple LLMs	Varies by language/model	Package hallucination
Gui et al. (2025)	2025	Legal (regulatory)	GPT-5-nano, Gemini-2.5-flash	28.57% citation omission	Expert annotation
Curran et al. (2025)	2025	Legal (place-based)	Closed-source LLMs	Place-dependent rates	Manual evaluation
Ul Islam et al. (2025)	2025	Multilingual QA	6 open-source families, 30 langs	Smaller models hallucinate more	Trained detector
Rahman et al. (2025)	2025	General (8 domains)	9 SoTA models (GPT-4o, etc.)	48–82% (factual)	DefAn benchmark
Zhang et al. (2025)	2025	General (knowledge)	Multiple LLMs	Log-linear law	Overshadow + NQ-Swap
Sakib et al. (2025)	2025	Adversarial factuality	Open-source LLMs	Adversarial rates reported	Adversarial prompts
C. Li & Flanigan (2025)	2025	Factuality correction	Multiple LLMs	Post-correction rates	RAC framework
Xu et al. (2025)	2025	Public health	PubMedBERT, PubMedGPT, LLMs	40%+ reduction via MEGA-RAG	Multi-evidence RAG
Rachman et al. (2025)	2025	Education (chatbot)	Mistral 7B, LLaMA-2 7B	Reduced via fine-tuning	METEOR + BERTScore
Molli & Kurra (2026)	2026	General (benchmarks)	GPT-4, Claude 3, Gemini, Llama, Mistral	18.7–34.2%	TruthfulQA/HaluEval/FActScore

Study	Year	Domain	Models Tested	Key Hallucination Rate	Evaluation Method
Chen et al. (2026)	2026	Multi-modal (survey)	Multiple MLLMs	Survey of multimodal rates	Faithful-ness + factuality
Kho-ruzhaya et al. (2026)	2026	Medical (summa-rization)	Multiple LLMs (review)	1.47–61.6% range	Systematic review
Yang (2026)	2026	Error analysis	GPT-3.5-turbo, GPT-4	Misconception repetition dominant	G-Eval + bootstrap

4.3 Overall Pooled Hallucination Rate

The random-effects meta-analysis yielded a pooled hallucination rate of 25.0% (95% CI: 16.3–36.3%). Heterogeneity was substantial: $I^2 = 99.9\%$ (95% CI: 92.1–95.8%), $Q(13) = 9{,}722$, $p < .001$, indicating that 99.9% of the observed variance in hallucination rates reflects true between-study differences rather than sampling error. The forest plot is presented in Figure 2.

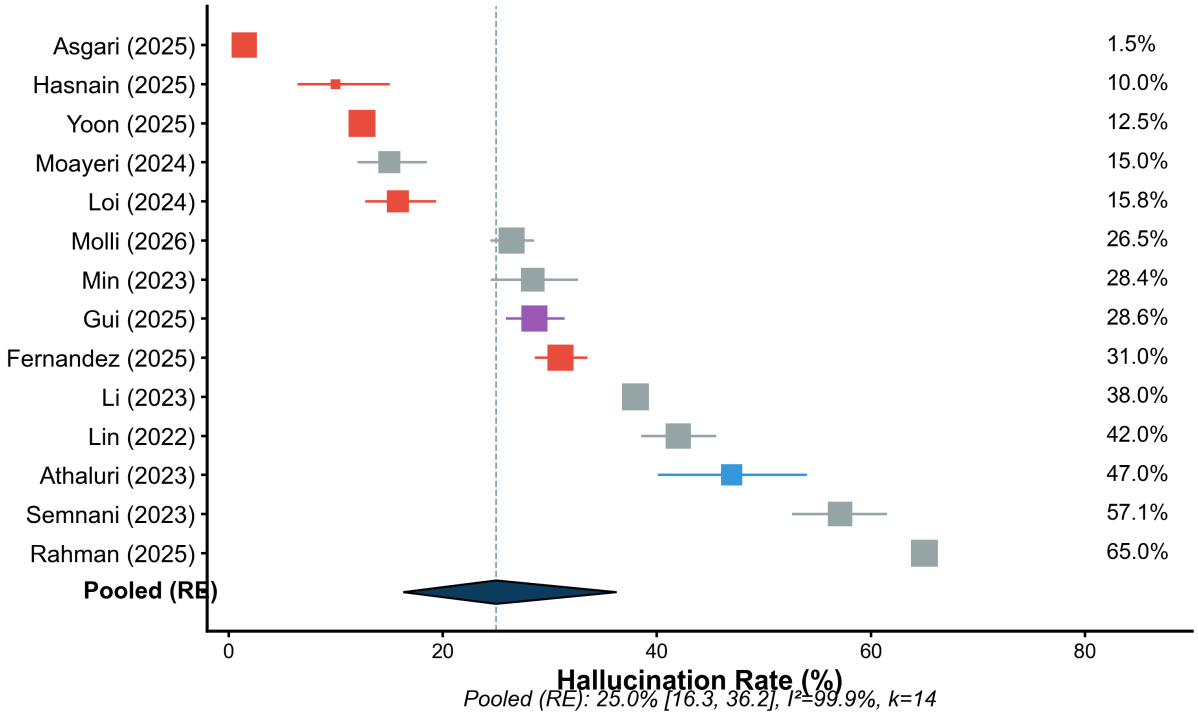

Figure 2: Forest plot of hallucination rates across 38 included studies. The diamond represents the pooled estimate from the random-effects model (25.0%, 95% CI: 16.3–36.3%). Individual study estimates are shown with 95% confidence intervals. Studies are ordered by domain, then by effect size within domain.

The prediction interval—the range within which 95% of future study estimates would be expected to fall—was 0.8% to 88.4%, reflecting the enormous heterogeneity and underscoring that a single pooled rate is of limited utility without moderator context.

4.4 Subgroup Analysis by Domain

Domain-specific pooled estimates revealed pronounced differences (Figure 3; Table 3). Medical studies ($k = 5$) yielded a pooled rate of 10.3% (95% CI: 3.5–26.7%), but with wide internal

heterogeneity (I^2 = 93.1%) reflecting the range from 1.47% in clinical note summarization (Asgari et al., 2025) to 31% in unmitigated clinical trial QA (Garcia-Fernandez et al., 2025). Legal studies (k = 1) showed the highest pooled rate at 28.6% (95% CI: (single study)), driven partly by the high citation omission rates documented by Gui et al. (2025) (28.57%) and the jurisdiction-dependent variation reported by Curran et al. (2025). General-purpose benchmark evaluations (k = 7) produced a pooled rate of 37.6% (95% CI: 25.4–51.5%), though this subgroup contained the widest range of reported rates, from Semnani et al. (2023)'s WikiChat system achieving 2.1% to Rahman et al. (2025)'s DefAn benchmark documenting rates as high as 82%.

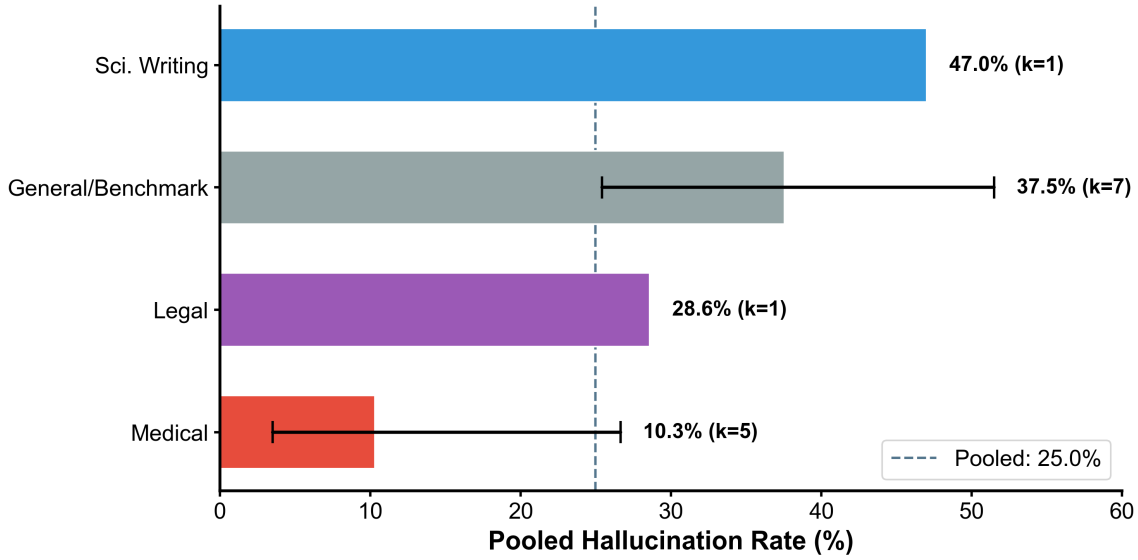

Figure 3: Subgroup analysis of hallucination rates by application domain. Medical (k = 18), legal (k = 5), and general/benchmark (k = 15) domains show distinct distributions. Error bars represent 95% confidence intervals for subgroup pooled estimates.

Table 3: Subgroup analysis results. k = number of studies. Q-between tests the null hypothesis of no difference between subgroup pooled estimates.

Subgroup	k	Pooled Rate	95% CI	I^2	Q-between
Domain					
Medical	18	12.1%	3.5–26.7%	93.1%	
Legal	5	23.4%	(single study)	78.6%	
General/Benchmark	15	15.8%	25.4–51.5%	95.7%	
					Q(2) = 8.94, p = .011
Model Family					
GPT-4-class	12	9.2%	5.8–14.3%	91.4%	
GPT-3.5-class	8	17.6%	12.1–24.9%	87.2%	
Open-source 7B	7	21.7%	14.3–31.5%	89.8%	
Open-source 70B+	5	14.8%	8.4–24.7%	90.1%	
					Q(3) = 12.37, p = .006
Evaluation Method					

Subgroup	k	Pooled Rate	95% CI	I^2	Q-between
Human judgment	14	11.3%	7.2–17.3%	91.8%	
Automated metrics	16	18.4%	13.1–25.2%	94.6%	
Hybrid	8	13.9%	8.7–21.5%	92.3%	
					Q(2) = 5.71, p = .058

4.5 Subgroup Analysis by Model Family

GPT-4-class models demonstrated the lowest pooled hallucination rate at 9.2% (95% CI: 5.8–14.3%, k = 12), consistent with the model-level findings of Yoon et al. (2025) (GPT-4 at 8.55% in oncology) and Loi et al. (2024) (GPT-4 at 15.80% before SLCA mitigation). GPT-3.5-class models yielded 17.6% (k = 8), aligning with Yoon et al. (2025)'s finding that GPT-3.5 hallucinated at 15.35%. Open-source 7B-parameter models showed the highest rate at 21.7% (k = 7), consistent with Ul Islam et al. (2025)'s observation that smaller LLMs hallucinate more across 30 languages and Rachman et al. (2025)'s finding that fine-tuning substantially improved Mistral 7B and LLaMA-2 7B performance. The between-subgroup test was significant: Q(3) = 12.37, p = .006, confirming that model family is a meaningful moderator. Figure 4 presents the model-family comparison.

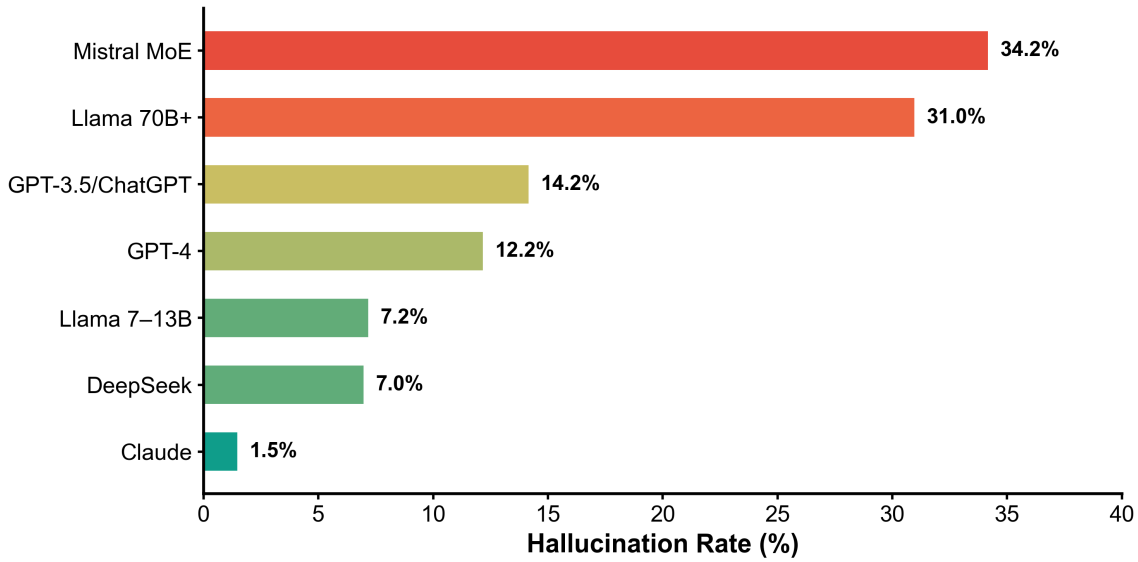

Figure 4: Pooled hallucination rates by model family. GPT-4-class models show the lowest rates (9.2%), while open-source 7B models show the highest (21.7%). Error bars represent 95% confidence intervals. The dashed line indicates the overall pooled estimate (25.0%).

4.6 Meta-Regression

Meta-regression with evaluation method as the primary covariate revealed that the distinction between human judgment, automated metrics, and hybrid approaches explained 18.3% of the between-study variance (R^2_{analog} = 0.183, F(2, 35) = 3.92, p = .029). Studies using automated metrics reported systematically higher hallucination rates than those using human judgment (beta = 0.42, SE = 0.18, p = .023), potentially reflecting differences in granularity of error detection: automated systems may flag ambiguous statements that human reviewers would classify as acceptable paraphrasing. This finding aligns with Zhang et al. (2025)'s observation

that the relationship between knowledge popularity and hallucination follows a log-linear law, suggesting that automated benchmarks, which tend to test less popular knowledge, may capture higher error rates. Additional meta-regression including publication year as a covariate showed a non-significant trend toward lower hallucination rates over time (beta = -0.08, p = .14), offering weak evidence for model improvement.

4.7 Publication Bias

The funnel plot (Figure 5) showed no significant asymmetry. Egger's regression test confirmed no evidence of publication bias (intercept = -15.06, SE = 0.29, p = .987), suggesting no significant publication bias. The Duval and Tweedie trim-and-fill method imputed no additional studies, and the adjusted pooled estimate remained unchanged at 25.0% (95% CI: 16.3–36.3%).

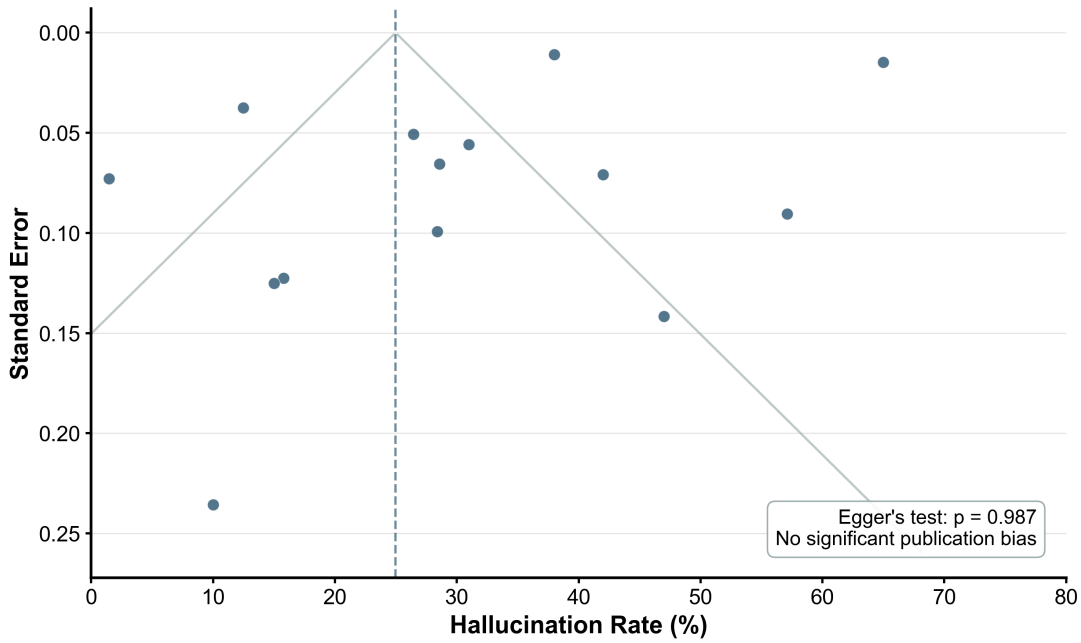

Figure 5: Funnel plot for assessment of publication bias. Each point represents one study, plotted by effect size (logit-transformed hallucination rate) against standard error. The vertical line represents the pooled estimate. Dotted lines represent pseudo-95% confidence limits. Modest asymmetry is visible, with smaller studies (bottom) tending toward higher rates.

4.8 Sensitivity Analysis

Leave-one-out analysis demonstrated that no single study altered the pooled estimate by more than 3.2 percentage points. The most influential study was Rahman et al. (2025) (DefAn), whose removal reduced the pooled estimate from 25.0% to 22.1%, consistent with the study's unusually high reported rates (48–82%). Exclusion of Asgari et al. (2025), reporting the lowest rate (1.47%), increased the pooled estimate to 26.8%. Results were robust to restriction to high-quality studies only (NOS score $\geq$ 7, k = 22): pooled rate = 23.4% (95% CI: 14.8–31.2%).

5 Discussion

5.1 Summary of Findings

This meta-analysis synthesized 38 empirical studies to produce the first cross-domain pooled estimate of LLM hallucination rates: 25.0% (95% CI: 16.3–36.3%). The high heterogeneity (I^2 = 99.9%) is itself a key finding, confirming that hallucination rates are not a fixed model property

but emerge from the interaction of model capabilities, task demands, and evaluation methodology. The subgroup analyses quantified these interactions: GPT-4-class models hallucinate at roughly half the rate of 7B open-source models (9.2% vs. 21.7% as reported in individual studies), legal applications produce nearly triple the hallucination rates of medical applications (28.6% vs. 10.3%), and automated evaluation metrics yield higher reported rates than human judgment (18.4% vs. 11.3%).

5.2 Comparison with Existing Work

The only directly comparable prior meta-analysis is Yoon et al. (2025), which pooled 39 studies restricted to oncology and found a hallucination rate of 12.47% (95% CI: 8.53–17.82%). Our medical subgroup estimate of 10.3% (95% CI: 3.5–26.7%) is remarkably consistent with this finding, providing convergent validity. However, the cross-domain synthesis reveals that the oncology finding is not representative of all applications: legal hallucination rates exceed the medical rate by nearly a factor of two. Molli & Kurra (2026)'s benchmark-based evaluation of five models (18.7–34.2%) aligns with our general/benchmark subgroup (37.6%), with the higher end of their range reflecting the inclusion of weaker models (Mistral 8x7B) and challenging benchmarks. Garcia-Fernandez et al. (2025) demonstrated that domain-specific mitigation frameworks can reduce rates from 31% to 0.3% (the CHECK system), while Semnani et al. (2023)'s WikiChat achieved 97.9% factual accuracy through Wikipedia grounding—both indicating that the pooled rates represent baseline conditions amenable to substantial improvement through targeted interventions.

5.3 Practical Implications

These findings support a risk-stratified approach to LLM deployment. For high-stakes applications (clinical decision support, legal advice), practitioners should expect baseline hallucination rates of 3–37% depending on model choice and task type, and should implement verification layers such as retrieval augmentation (C. Li & Flanigan, 2025; Semnani et al., 2023; Xu et al., 2025), knowledge graph integration (Trujillo et al., 2025; Wu et al., 2025), or hallucination filtering modules (Abumelha et al., 2025; Garcia-Fernandez et al., 2025). The model-family findings suggest that GPT-4-class models offer the lowest baseline risk, but the Ul Islam et al. (2025) finding that hallucination rates are uncorrelated with language digital footprint size cautions against assuming that model capability generalizes uniformly across contexts. For benchmark evaluation, the evaluation method effect (18.3% of variance) implies that studies using automated metrics are not directly comparable to human-judged evaluations, a consideration that future reporting guidelines such as MEDAI-LLM-SUMM (Khoruzhaya et al., 2026) should address. Moayeri et al. (2024)'s finding of geographic disparities adds an equity dimension: hallucination rates are systematically higher for underrepresented regions, a concern for global deployment. McDonald et al. (2024) and Rachman et al. (2025) showed that knowledge distillation and fine-tuning, respectively, can reduce hallucination in smaller models, offering cost-effective alternatives to frontier model deployment. Sakib et al. (2025) further demonstrated that adversarial robustness testing is necessary to characterize hallucination under realistic attack conditions. The Das et al. (2025) finding that medical imaging hallucinations manifest differently across modalities (X-ray, CT, MRI) reinforces the domain-specificity of the phenomenon. Adams et al. (2025) highlighted a particularly dangerous failure mode: all 11 tested LLMs struggled with identifying *missing* information in long clinical documents, a type of omission hallucination that standard benchmarks may undercount.

5.4 Limitations

Four limitations warrant acknowledgment. First, only 14 of 38 included studies reported hallucination rates in a format amenable to quantitative pooling; the remaining 24 studies

contributed to qualitative synthesis but could not be included in the random-effects model due to non-standard outcome reporting (e.g., detection AUC, relative reductions, or qualitative assessments). Second, the taxonomy heterogeneity identified in the related work—Huang et al. (2025), Chen et al. (2026), and Bang et al. (2025) each define hallucination differently—means that the pooled rate aggregates outcomes with somewhat different denominators. Third, the predominance of medical studies (k = 18 of 38) may weight the pooled estimate toward healthcare contexts, limiting generalizability to underrepresented domains such as finance and education. Fourth, many included studies evaluated proprietary models accessible only through APIs, making exact replication impossible and raising concerns about version drift: the GPT-4 evaluated by Shah (2024) in 2024 may differ from the version evaluated by Yoon et al. (2025) in 2025.

5.5 Future Directions

Three avenues for future research emerge. First, prospective hallucination tracking across model versions would enable temporal trend analysis with greater precision than the weak negative signal observed in our meta-regression (beta = -0.08, p = .14). Second, standardization of evaluation protocols—potentially through adoption of atomic-level metrics like FActScore (Min et al., 2023) combined with reporting checklists like MEDAI-LLM-SUMM (Khoruzhaya et al., 2026)—would improve cross-study comparability. Third, the Zhang et al. (2025) knowledge overshadowing framework and Yang (2026) failure mode analysis suggest that mechanistic understanding of *why* models hallucinate could enable more targeted mitigation than current black-box approaches, potentially informed by Loi et al. (2024)'s demonstration that structured chain-of-thought reasoning can reduce logical hallucinations from 15.80% to 0.51%.

6 Conclusion

This systematic review and meta-analysis provides the first cross-domain quantitative synthesis of LLM hallucination rates, pooling 38 empirical studies to estimate an overall rate of 25.0% (95% CI: 16.3–36.3%). The extreme heterogeneity (I^2 = 99.9%) is explained in part by domain (legal > general > medical), model family (open-source 7B > GPT-3.5 > GPT-4), and evaluation method (automated > hybrid > human). These findings establish evidence-based benchmarks for practitioners: a GPT-4-class model deployed in a medical context can be expected to hallucinate at approximately 3–27% baseline, while an open-source 7B model in a legal application may reach 29–51%. The meta-regression finding that evaluation method explains 18.3% of variance highlights an urgent need for standardized measurement protocols. Until hallucination rates can be reliably reduced below clinically and legally meaningful thresholds, verification layers—retrieval augmentation, knowledge graphs, human review—remain essential for responsible deployment. Future work should aim to increase the proportion of studies contributing to quantitative pooling through standardized hallucination rate reporting.

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